Experiment 21: Write a Prolog Program to Implement Towers of Hanoi.

Aim: To implement the Towers of Hanoi problem using recursion in Prolog.

Procedure:

1. Open Notepad and write the Prolog program.
2. Define the base case for moving one disk.
3. Define the recursive rule for moving disks.
4. Move the disks to the auxiliary peg.
5. Move the largest disk to the destination peg.
6. Display all the required moves.

AI_exp21.pl

File Edit View

```
hanoi(1, Source, Destination, _) :-
    write('Move disk from '),
    write(Source),
    write(' to '),
    writeln(Destination).

hanoi(N, Source, Destination, Auxiliary) :-
    N > 1,
    N1 is N - 1,
    hanoi(N1, Source, Auxiliary, Destination),
    hanoi(1, Source, Destination, Auxiliary),
    hanoi(N1, Auxiliary, Destination, Source).
```

Output:

```
11 ?- consult('C:/Users/acer/Downloads/AI_exp21.pl').
true.

12 ?- hanoi(3, a, c, b).
Move disk from a to c
Move disk from a to b
Move disk from c to b
Move disk from a to c
Move disk from b to a
Move disk from b to c
Move disk from a to c
true
```

Result: Thus, the Towers of Hanoi problem was implemented successfully using Prolog recursion.